

WMS Build Roadmap

as of 2026-09-14

Warehouse Management System — build pipeline snapshot

Master Data → Yard & Gate → Inbound → Putaway → Inventory → Outbound → Picking → Dispatch → Returns → Analytics

● Built ● Started ● Not started



What's landed since the last refresh (2026-09-14)

The next real module in the build order — Outbound + Picking's backend and frontend are both built and live-verified; a competitor gap-analysis pass followed the same day.

Outbound Orders + Picking — backend built & verified

New **OutboundOrder/OutboundOrderLine/PickTask/PickTrip** schema. Orders are uploaded vehicle-wise but the vehicle is resolved *after* creation — matched against an already-gated-in vehicle by destination city, then a Supervisor+ explicitly allots one. Allotment reserves real pick tasks against actual on-hand, position-scoped (never a lane-combined total — a real bug was found and fixed here). Picking itself: FIFO claiming (no barcode), two-tier scan completion (pallet scan for a whole-unit pick, per-unit barcode scans for a partial one), real PICK/PICK_IN ledger writes. 32/32 diagnostic checks passed.

Picking gap analysis vs. other WMS platforms

A research pass against Manhattan Active WM, SAP EWM, and Blue Yonder, cross-checked directly against our own code (not just design docs) — see the dedicated table on page 5. Found two real bug-class gaps (no claim-expiry for an abandoned pick trip; a stuck-forever NEEDS_SOURCE task with no retry path) plus three integration/capability gaps (Pick Face not yet consulted by Picking, no operator-fairness layer, no Picking metric in Analytics).

Outbound Orders + Picking — frontend built & verified

New **Outbound Orders** page (order maker + Allot-Vehicle modal) and **Picking** page (Claim Next Task, two-tier completion form, task queue). Verified live through the real rendered UI end to end: created a real order, allotted a real gated-in vehicle, claimed and completed a real pick trip via barcode scan, confirmed the order flipped to PICKED with the right quantities.

Next up

Dispatch is the one piece left in this trio — the isPriority "can't be missed" gate, the real DISPATCH ledger write, and releasing the dock/vehicle/pallets. Needs its own short align-before-coding pass, plus a decision on whether to close the two real Picking gaps first (see page 5).

Module-by-module status

Module	Status	Summary
Master Data	Built	Warehouses (network-node master: Factory/DC/CNF/Cross-dock), SKUs, Customers, Locations (with a full 3D Plan View), Users — all with role-scoped access and standard import/export.
Yard & Gate	Built	Gate In/Out, dock assignment, document checks, real compass-direction dock zones + a live-rotating 3D compass overlay. One competitor-research pass done (self-check-in + physical inspection flagged, not built).
Inbound	Built	Order maker + Excel bulk import + ERP push, vehicle-matched orders, scan-based receiving, Complete Inward Process/Dock Out, per-dock default staging locations.
Putaway	Built	Real bin-suggestion logic for every storage type (Rack/SPR, Drive-in, Ground/Floor, Stillage) — ABC/FMS-aware, reservation-aware, dock-relative, and (Rack + Stillage) row/rank-aware. Pallet consolidation, Pick Face for SPR, operator-assignment fairness, Aisle/Level Rank visuals. Open: cancel path, queue ordering, Pick Face for Drive-in/Ground/Stillage, Stillage's "closed column" lifecycle tracking.
Inventory	Built	On-hand Line Items + SKU Summary views, a transaction-level ledger, and Excel export on all three views (with From/To Location) — all live-derived, no stored counters. The Daily Volume dashboard's outward half is a separate, still-open Analytics item, unblocked now that Picking exists.
Outbound	Started	Order maker built (manual creation, candidate-vehicle matching by destination city, Supervisor+ allotment reserving real pick tasks at real on-hand). Not built: Excel import, ERP push, vehicle weight/volume capacity check.
Picking	Started	Core execution built: position-scoped dynamic source resolution (lane-aware, front-most, aging-bucket FIFO), FIFO trip claiming, two-tier scan completion, real PICK/PICK_IN ledger. Real gaps found in a same-day competitor review — see page 5. Wave/batch/zone/cluster modes and Task Interleaving with Putaway still deliberately deferred.
Dispatch	Not started	Schema exists (isPriority gate, Dispatch model). No design conversation has happened yet — the next real piece of this trio.
Returns	Not started	Nothing built yet.
Analytics	Started	Operator productivity (marrying/putaway time per operator, Pick Face trip time, abandoned-claim flags) plus Daily Inward Volume charts. Missing a Picking metric — flagged on page 5.

Immediate candidates for the next session

Live options on the table — not a forced order.

1. **Dispatch** — the last piece of the Outbound/Picking/Dispatch trio. Needs its own align-before-coding pass: whether an order can dispatch with non-priority lines still short as long as priority lines are complete, what the Dispatch action itself is (a new explicit action vs. piggybacking on the existing Gate Out step), and whether load needs its own re-scan or can trust the picking ledger.
2. **Close the two real Picking gaps** (found this session, see page 5) — no claim-expiry for an abandoned PickTrip (Putaway already solved this exact problem), and a NEEDS_SOURCE task has no retry path at all (Putaway's "Request Different Bin" has no Picking equivalent). Both are bug-class omissions, not scope decisions.
3. **Wire Picking into Pick Face** — suggestPickPosition() is scoped to ACTUAL_STORAGE only; the whole reason Pick Face exists (fast, easy access to prioritized stock) goes unused by Picking today.
4. Operator-assignment fairness for Picking — Putaway has a full idle-time-ranking + Supervisor/Manager escalation layer (2026-09-02); Picking has plain FIFO claiming and nothing else.
5. Add a Picking metric to Analytics — Marrying/Putaway/Pick Face are all reported per operator; PickTrip isn't.
6. The reslotting/consolidation suggestion engine — genuinely unblocked (ABC, dock-relative placement, and FMS are all in place). Schema has sat ready since Topic 1.
7. Daily Outward Volume (Analytics) — the mirror of the already-built Daily Inward Volume chart. Now genuinely unblocked (Picking writes real PICK movements) — still needs a design conversation on exactly which movement(s) represent the warehouse-exit event.
8. Exception/short-pick handling mid-pick — every WMS surveyed lets an operator flag "not actually there"/damaged at the point of picking, routing to a Supervisor (mirrors Inbound's own blocked-scan/approve pattern). Not designed yet for Picking.
9. Pick Face for Drive-in / Ground / Stillage — SPR-only today, deliberately.
10. Dock-out → Gate-out signal, Dock + Staging + Yard visualizer, a cancel/void path for a mistaken Gate In — all previously flagged, still not designed.

Deferred, lower priority — per explicit client calls, don't build unprompted

From the top-WMS gap checks (Yard/Gate 2026-09-07, Picking 2026-09-14 — see page 5)

- **Wave / batch / zone / cluster picking** — you confirmed "basics first"; single-task discrete picking only, for now.
- **Task Interleaving** — the one exception on this list: it's flagged HIGH PRIORITY now that Picking genuinely exists, not just "someday."
- Cycle counting / physical inventory reconciliation — no workflow exists at all yet.
- 3PL / client billing — a real gap against the multi-company SaaS goal; no rate cards or invoicing today.
- Vendor/Supplier master data — receiving is tied to a Vehicle, not to a vendor.
- Kitting/Assembly logic — schema exists, zero logic built.
- Quality Control / Inspection workflow — no formal hold→inspect→release/reject module.
- Cartonization / pack-station / sortation — likely an e-commerce-parcel concept; may not apply to our pallet/case B2B model. Flagged, not assumed either way.
- Voice picking / pick-to-light — hardware investments, not software gaps.

Putaway follow-ons

- A cancel/exception path for a task that can't be completed.
- Correcting an already-completed mis-putaway.
- Real queue-ordering / aging-based task prioritization.
- Pick Face for Drive-in / Ground / Stillage (SPR-only today, deliberately).
- Stillage's "closed column" lifecycle tracking — unblocked now that Picking writes real `MovementType.PICK` movements, not yet picked up.
- Pallet reuse-on-depletion — now genuinely exercisable (Picking writes `PICK`), not yet wired/tested.

Everything else explicitly parked

- Self-service driver check-in, Yard Plan View, closing a short receipt, Batch/Lot + expiry capture.
- Real GS1/barcode parsing, camera-based scanning, real SMS/Email/WhatsApp/Voice providers.
- Dock Scheduling algorithm, blacklist enforcement at Gate In, WMS/TMS integration.
- `SUPER_ADMIN` creation, a configurable permission matrix, an invite-link flow.
- Inventory Control Policy, Opening Balance load, dispatch-proximity distance calc, real SAP/ERP integration, cloud deployment.

Picking — gap analysis vs. other WMS platforms (2026-09-14)

Researched against Manhattan Active WM, SAP EWM, and Blue Yonder, then checked directly against our own code (not just the design doc) — each row below is either a confirmed real gap in what's built, a capability that needs its own design pass, or something already deliberately deferred by your own call.

Gap	Type	What other WMS platforms do	Status in our system	Priority
No claim-expiry for an abandoned pick trip	Real gap	An RF/scan-driven claim that's never completed times out and returns to the pool automatically.	Putaway already solved this exact problem (PutawayClaimExpiryScheduler, 30-min auto-expiry). Picking has no scheduler at all — a claimed-then-abandoned PickTrip stays IN_PROGRESS forever, permanently blocking that task.	High
NEEDS_SOURCE task has no retry path	Real gap	A short-stock/no-source condition routes to an alternate-location search or a Supervisor exception queue.	Putaway's equivalent (NEEDS_BIN) has a real "Request Different Bin" action that re-runs the suggestion. Picking's NEEDS_SOURCE has zero resolution path anywhere — a genuine stockout at allotment creates a permanently dead task with no visibility to anyone.	High
Picking doesn't use Pick Face	Real gap	Fast-access forward-pick locations are always preferred over reserve storage for an eligible SKU.	suggestPickPosition() is scoped to ACTUAL_STORAGE only — its own code comment already flags this as a follow-on. Pick Face (built for SPR, 2026-09-05) sits completely unused by Picking; an order for a Pick-Face-eligible SKU still always routes to reserve.	High
No operator-fairness layer for Picking	Real gap	Idle-time-based task assignment recommendations, with supervisor escalation if work backs up.	Putaway has a full layer (2026-09-02): idle-time ranking, priority-task surfacing, two-tier Supervisor→Manager escalation. Picking has plain FIFO claiming and nothing else.	Medium
No Picking metric in Analytics	Real gap	Per-operator trip time / productivity reported for every pick-type activity.	AnalyticsService reports Marrying, Putaway, and Pick Face operator productivity — nothing for PickTrip. No abandoned-claim list either (meaningful only once claim-expiry exists).	Medium
No mid-pick exception handling	Needs design	An operator can flag "correct location, item genuinely not there" or "damaged," routing to a Supervisor — same shape as a blocked Inbound scan.	A wrong-location scan hard-blocks; there's no "I'm at the right place and it's genuinely missing/damaged" path at all. No reason-code capture anywhere in Picking.	Medium
Wave / batch / zone / cluster picking	Deferred	Industry-standard picking orchestration — batching/waving orders together cuts travel 30–60% at scale.	Explicitly deferred by you — "basics first." Today's model is single-order, single-task discrete picking only.	Confirmed deferred
Task Interleaving (Putaway + Picking)	Deferred	Combines a putaway trip and a pick trip into one path to cut empty travel.	Already flagged HIGH PRIORITY in this roadmap for "the moment Picking exists" (2026-09-07) — that moment is now. Still not designed.	High (revisit now)
Cartonization / pack-station / sortation	Flag only	Optimal box selection and multi-item order consolidation before shipping — common in e-commerce-parcel WMS.	Not built, and may not apply — our model is pallet/case-level B2B fulfillment with a fixed per-order staging bin, not parcel consolidation. Worth a quick confirm, not assumed either way.	Confirm applicability
Voice picking / pick-to-light	Hardware	Hands-free picking technology — 25–50% productivity gains, >99% accuracy in mature deployments.	Not applicable to the software layer — a hardware investment decision, not a code gap. Flagged for awareness only.	Out of scope

Sources: Manhattan Active WM (manh.com), SAP EWM Picking Strategies (help.sap.com), Blue Yonder WMS (info.blueyonder.com), RFgen Warehouse Picking Guide, Lucas Systems Order Picking Productivity, Cleverence Warehouse Replenishment.